Combination circuits

antocciphysics | TEACHER KEY | CPA + MATCHING SEI

Learning target: Reduce a circuit by identifying genuine series and parallel groups.

Accept equivalent correct methods and clearly labeled models. Language supports do not change the physics demand. Supplied data are examples; actual lab measurements and teacher checkpoints must be recorded from the lesson.

01. Warm-up: bring back an earlier idea *Review: Unit 7, Day 6.*
V=IR=15 V.

02. Worked example: explain the model
Series resistor drop 6 V; parallel section 6 V; each 4 ohm branch 1.5 A.

03. Read and interpret the model
One resistor before a split into two 6 Ω branches, then return to supply.

04. Apply the physics
5 Ω total; 2 A.

05. Explain your reasoning
Electrical connections and shared nodes, not page position.

06. Review: connect an earlier idea *Review: Unit 6, Day 15.*
3000 m; range is half the full path.

07. Exit check
1 A each.

08. STRETCH: test the model
Parallel 4 ohms; total 6 ohms; current 2 A.

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01. Warm-up: bring back an earlier idea *Review: Unit 7, Day 6.*
R=30 Ω; V/I is constant at these points under the measured conditions.

02. Worked example: explain the model
Series resistor drop 6 V; parallel section 6 V; each 4 ohm branch 1.5 A.

03. Read and interpret the model
One resistor before a split into two 6 Ω branches, then return to supply.

04. Apply the physics
$R_{parallel}$=3 Ω; total 5 Ω; I=2 A; series drop 4 V; branch voltage 6 V; each branch 1 A.

05. Explain your reasoning
Electrical connections and shared nodes, not page position.

06. Review: connect an earlier idea *Review: Unit 6, Day 15.*
At the same wave speed, equal range gives equal out-and-back time. Doppler shift depends on radial velocity, so motion can differ.

07. Exit check
Parallel equivalent 2 Ω; total 4 Ω; current 2.5 A.

08. STRETCH: test the model
Parallel 2, total 5 ohms; total 3 A; parallel voltage 6 V; branches 1 and 2 A.